Prodigy Anesthesia Quick Review

Laminectomy

What is a laminectomy and why is it performed?	A laminectomy is the complete removal of the vertebral lamina in order to decompress the neural elements of the spinal cord. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 122.*
What is the difference between a laminotomy, a laminectomy and a decompressive laminectomy?	Lumbar laminotomy is the removal of a part of the lamina. Laminectomy is complete removal of the lamina. Both can be used to treat radiculopathy due to degenerative disc disease. A decompressive laminectomy is used to treat compression of the cauda equina due to degenerative disc disease, spinal stenosis, neoplasm, trauma, or stenosis. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 122.*
How can an anesthetist assist the surgeon in evaluating for a dural tear following a laminectomy?	By performing a Val Salva maneuver (sustained inspiratory pressure of 30-40 cm H2O) to test the integrity of the dura. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 122.*
What structures are at risk for damage during a lumbar laminectomy?	Because the surgery is performed via a posterior approach, the retroperitoneal structures such as intestines, gastrointestinal structures, and great vessels are at risk for damage. The dura and spinal cord are also at risk for damage during this procedure. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 122.*
In what positions are lumbar microdiscectomies typically performed? How is this related to the risk of bleeding?	Lumbar microdiscectomy may be performed in the prone or kneeling positions, being careful to allow the abdomen to hang free to prevent venous congestion in the epidural veins which increases bleeding. *Jaffe RA, Samuels SI. Anesthesiologist's Manual of Surgical Procedures. 4th ed. Philadelphia, PA: Lippincott Williams and Wilkins, 2009: 962-963.*
What forms of anesthesia are acceptable for lumbar microdiscectomy?	Spinal anesthesia or general anesthesia are both acceptable for traditional lumbar microdiscectomy. Percutaneous microdiscectomies (minimally-invasive) are typically performed under sedation (MAC) with local anesthesia. *Jaffe RA, Samuels SI. Anesthesiologist's Manual of Surgical Procedures. 4th ed. Philadelphia, PA: Lippincott Williams and Wilkins, 2009: 962-963.*

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What is the most common complications from using the Andrews frame for performing a lumbar laminectomy?	The Andrews frame secures the patient in knee-chest position with the abdomen hanging freely which reduces intra-abdominal pressure and contributes to decreased blood loss. Because the legs are below the level of the heart, venous return is decreased and severe hypotension can result. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1009.*
In what ways can the patient be positioned for a posterior cervical laminectomy? What are the anesthetic implications of positioning for this procedure?	Although the sitting position is becoming more rare, it is still performed by surgeons who prefer it over the prone position. A decompressive laminectomy is performed via a posterior, midline approach with the head secured either in pins or a horseshoe headrest. Posterior procedures performed in the sitting position place the patient at risk for air embolism. It is prudent in these cases to monitor heart sounds with a precordial Doppler and place a central venous line prior to surgery in order to aspirate air should an embolism occur. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 111-112.*
What are the potential complications of an anterior cervical laminectomy and/or discectomy?	Complications from anterior cervical surgery include esophageal perforation, venous air embolus, damage to the recurrent laryngeal nerve from retractors, tension pneumothorax, and airway obstruction from hematoma. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 112-113.*

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